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Class 9 Biology

Chapter 9

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Unit no. 9

Topic: Plant Reproduction

1. Define Asexual Reproduction and Describe different types of asexual production?

Asexual Reproduction: In this process, only one parent is involved and it gives rise to offspring that are essentially clones sharing the exact genetic makeup of the parent.

Types of Asexual Reproduction: Following are the types of Asexual Reproduction includes:

- Binary Fission
- Budding
- Spore Formation
- Vegetative propagation

Binary Fission:

The term "fission" means "to divide". During binary fission, the parent cell divides into two cells. Amoeba and Euglena exhibit binary fission.

Binary Fission in Bacteria:

In bacteria, the parent cell divides into two, each daughter cell carrying a nucleus of its own that is genetically identical to the parent. The cytoplasm also divides leading to two equal-sized daughter cells. The process repeats itself and the daughter cells grow and further divide.

Budding

Budding is the process of producing an individual through the buds that develop on the parent body. Hydra is an organism that reproduces by budding. The bud derives nutrition and shelter from the parent organism and detaches once it is fully grown.

Spore Formation

Spore formation is another means of asexual reproduction. In Rhizopus, during unfavourable conditions, the organism develops sac-like structures called sporangium that contain spores. When the conditions are favourable, the sporangium bursts open and spores are released that germinate to give rise to new organisms.

Exospores: In actinomycete, spores are formed outside the bacterial cells, so they are called exospores.

Endospores: In Clostridium and Bacillus species, spores are formed inside bacterial cell, so are called endospores.

Parthenogenesis:

Parthenogenesis is the development of an offspring directly from unfertilized eggs. In flowering plants, it is defined as the asexual formation of a seed from the maternal tissues of the ovule, avoiding the processes of meiosis and fertilization, leading to embryo development. It is also known as apomixis.

Vegetative Propagation

Asexual reproduction in plants occurs through their vegetative parts such as leaves, roots, stems, and buds. This is called vegetative propagation. For example, potato tubers, runners/stolon, onion bulbs, etc., all reproduce through vegetative propagation

Kinds of vegetative propagation: there are two kinds of vegetative propagation:

- Natural vegetative propagation (plants do their own)
- Artificial vegetative propagation (people help with when needed)

2. Describe the methods of natural vegetative propagation in plants.

Ans: Following are the methods of natural vegetative propagation in plants:

- Bulbs
- Corm
- Rhizome
- Stem Tubers
- Stolon or Runners
- Suckers
- Vegetative propagation by leaves

Stem Tubers

Tuber of potato is an underground stem which stores food. It has buds in the depression called eye which can give rise to a new plant.

Rhizome:

Rhizome is an underground stem that have small scale like leaves. Along rhizome, there are distinct nodes and internodes that bears buds. Ex ginger, turmeric.

Bulb:

Bulb is a small disk-like stem surrounded by scale leaves. It is found in onion. In this, a new plant can arise from the stem.

By leaves:

Leaves of bryophyllum bear adventitious buds in the Notches in the margin of the leaf. When leaf follows the ground beef buds grow to form the independent plant.

Stolon/Runner:

Stolons, or runners, are above or below ground stem-like structures that connect one plant to another. Adventitious roots and a new plant can form from stolons. You can see this if you grow strawberry plants.

Suckers:

A sucker is any branch that grows out of the base of a bush or from an adventitious root. Once that branch is independent, the plant has propagated. Raspberries and lilac bushes behave in this manner.

Corm:

Corms are storage structures of some plants. They are swollen underground stems with dry-scale leaves covering them. In favorable conditions, they grow in size and can remain dormant in adverse conditions. Corms can be cut into pieces and each piece planted to produce a new plant. Dasheen and garlic are great examples of plants with corms.

3. Describe any two methods of artificial vegetative propagation in plants.

Following methods are used for artificial vegetative propagation in plants.

Cutting:

When we plant any part of the plant like a strip or stem by cutting it, then it is called harvesting. Sometimes cuttings are also used for root development. Cutting method is an important method of asexual reproduction. Because this method is asexual, therefore the plant produced by this method is genetically identical to its parents. Which is called a clone in common language.

Examples of Cutting: Rose, Bougainvillea chrysanthemum plants are propagated by this method.

Grafting:

Grafting is a method in which two plants are combined to form a new variety. In this method, any two varieties of the same crop are combined to form a new variety. In this method, two plants parts are involved in grafting:

Scion: The scion is the upper parts of the plant, which includes the desired characteristics such as specific fruit or flower traits.

Rootstock: The rootstock is the lower part , which provides the root system and stem support.

The scion is carefully attached or inserted into the rootstock.

Examples of Grafting: Mango, Guava varieties, Hibiscus, Roses, Bougainvillea varieties.

Tissue Culture: This procedure entails cultivating plant cells obtained from various regions of the parent plant. In plant tissue culture, a part of the plant such as a stem, leaf, embryo, anther, or seed can be used. The plant material is thoroughly sterilized using a combination of chemical treatments standardized for that species. Under sterile conditions, the plant material is placed on a plant tissue culture medium called agar that contains all the minerals, vitamins, and hormones required by the plant. The plant part often gives rise to an undifferentiated mass, known as a callus, from which, after a period of time, individual plantlets begin to grow. These can be separated; they are first grown under greenhouse conditions before they are moved to field conditions

Forexample: Example of tissue culture are oil palm, plantain, pine, banana, date, eggplant, jojoba, pineapple, rubber tree, cassava, yam, sweet potato, and tomato etc.